An Attack-Agnostic Defense Framework Against Manipulation Attacks under Local Differential Privacy

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***Abstract*—Protection of *local differential privacy* (LDP) protocols against *manipulation attacks* is an important and challenging problem. We hope to design an *attack-agnostic* framework, which does not rely on any knowledge of attackers. An early work [1] restricts the attacker's capability by converting each sample into a binary signal. However, the compression of signal leads to severe loss of information, and thus results in unnecessary sacrifice of utility, especially when $\epsilon > 1$. In this paper, we propose a general estimation framework RobustLDP for robust estimation under LDP. The general idea is to send carefully crafted pre-defined information to all users, and then aggregate the feedback at the server. We strike a better tradeoff between preserving information and restricting the attacker's capability. We instantiate RobustLDP for frequency estimation and mean estimation in ℓ_1 and ℓ_2 support, which serve as building blocks for more advanced tasks. We also establish theoretical guarantees for all possible attacks. The result shows that our method significantly outperforms the existing one for $\epsilon > 1$. Extensive experiments on multiple real-world datasets validate the effectiveness of our method.**

1. Introduction

Local differential privacy (LDP) has become the de facto standard for sensitive user data collection [2]. The general idea of LDP is to allow individual users to perturb their raw sensitive data before uploading it to the server. It ensures that attackers with access to the perturbed outputs cannot recover the raw data while guaranteeing the aggregator's ability to extract useful statistical information from the whole population. As such, LDP has been adopted by a number of high-tech giants. For example, Google integrated RAPPOR into the Chrome browser [3], Apple deployed LDP protocols in iOS [4] to collect user preferences, Microsoft also developed LDP protocols to collect application usage data [5].

Most existing studies on LDP assume that all the data providers are trustworthy and honestly report data following the LDP protocols [6], [7]. However, some data providers can be malicious in the real world. For instance, in Google, Twitter, and Hotmail, attackers can either compromise a number of existing accounts or purchase many new fake accounts from the underground market at very low prices [8].

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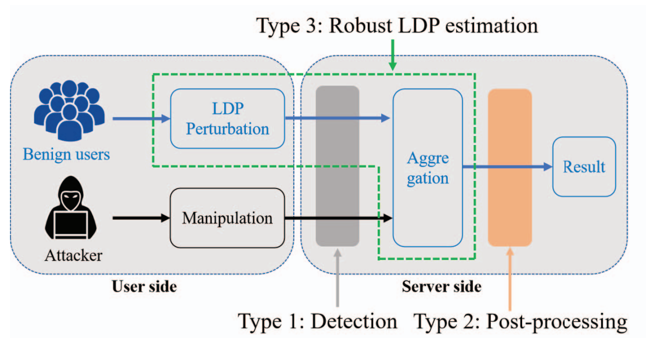

Figure 1: Illustration of three types of defense mechanisms against manipulation attacks under LDP.

These corrupted local data providers can report carefully designed data to the server, aiming to skew the final estimates, referred to as *manipulation attack*. Therefore, it is important to design effective defense strategies that can withstand these manipulation attacks.

Existing Defenses. Recent years have witnessed several defense mechanisms to mitigate the manipulation attacks [1], [9], [10], [11]. These strategies can be roughly categorized into three types: detection, post-processing, and robust estimation, as illustrated in Figure 1.

The *detection-based* methods aim to identify possibly corrupted reported data and remove them before aggregation, such as malicious user detection (MUD), conditional probability-based detection (CPAD) [10], and LDPGuard [9]. However, these methods rely on some information that is unlikely to be known in advance, such as the attack strategy and the number of corrupted samples. Moreover, according to recently established theories on robust statistics [12], [13], even under non-private setting, an attacker can corrupt a sample by roughly $\Theta(\sqrt{d})$ without being detected. Under LDP, the undetectable manipulation can become more serious. The *post-processing-based* methods aim to correct the aggregation results if they appear anomalous, such as LDPRecover [11]. However, these defense mechanisms can be easily bypassed. For instance, for the frequency estimation problem, the attacker can refine its attack to ensure that the final estimated values are nonnegative in all components and sum up to 1. Such an attack is unlikely to be corrected by post-processing based methods.

Both the detection-based and post-processing-based

2025 IEEE Symposium on Security and Privacy (SP) | 979-8-3315-2236-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/SP61157.2025.00255

DOI 10.1109/SP61157.2025.00255